8.1.2

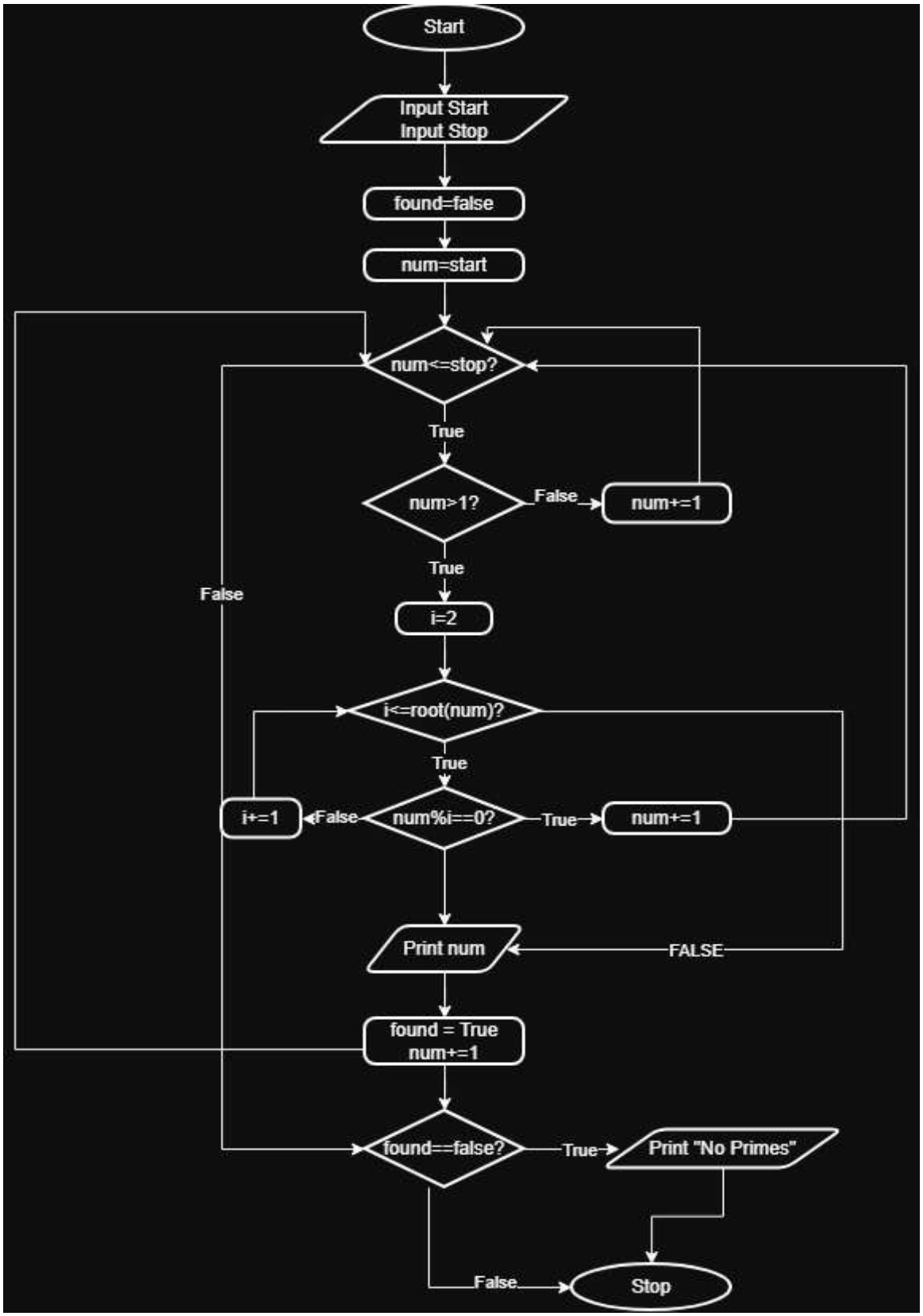
Problem Statement

Write a Python program to print all prime numbers within a given range (inclusive of both start and end values).

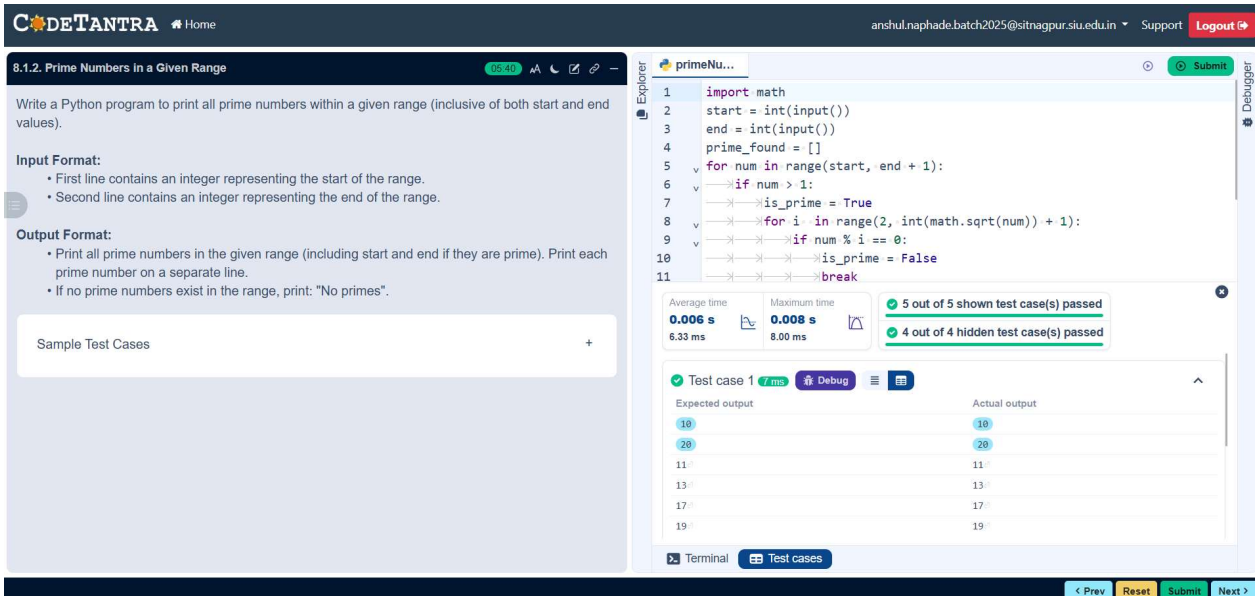
Algorithm

1. Start
2. Input the values *start* and *stop*
3. Set `found = False`
4. Repeat for each number *num* from *start* to *stop*
5. Check if *num* > *1*
6. Repeat for *i* from 2 to $\sqrt{num}$
7. If *num % i* == *0*, then break (number is not prime)
8. If no divisor is found:
    - Print *num*
    - Set `found = True`
9. After the loop, check `found`
10. If `found == False`, print **"No primes"**
11. Stop

Flowchart:


Start
Input Start
Input Stop
found=false
num=start
num<=stop?
True
num>1?
False
num+=1
True
False
i=2
i<=root(num)?
True
i+=1
False
num%i==0?
True
num+=1
Print num
FALSE
found = True
num+=1
found==false?
True
Print "No Primes"
False
Stop

EXECUTION:


CODETANTRA
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8.1.2. Prime Numbers in a Given Range
05:40
Write a Python program to print all prime numbers within a given range (inclusive of both start and end values).
Input Format:
• First line contains an integer representing the start of the range.
• Second line contains an integer representing the end of the range.
Output Format:
• Print all prime numbers in the given range (including start and end if they are prime). Print each prime number on a separate line.
• If no prime numbers exist in the range, print: "No primes".
Sample Test Cases
Explorer
primeNu...
Submit
Debugger
import math
start = int(input())
end = int(input())
prime_found = []
for num in range(start, end + 1):
if num > 1:
is_prime = True
for i in range(2, int(math.sqrt(num)) + 1):
if num % i == 0:
is_prime = False
break
Average time
0.006 s
6.33 ms
Maximum time
0.008 s
8.00 ms
5 out of 5 shown test case(s) passed
4 out of 4 hidden test case(s) passed
Test case 1
7 ms
Debug
Expected output
Actual output
10
20
11
13
17
19
Terminal
Test cases
Prev
Reset
Submit
Next